Qingxiang Guo, Ph.D.

Postdoctoral Scholar, Feinberg School of Medicine, Northwestern University
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I. Education

Ph.D., Huazhong Agricultural University, Wuhan, Hubei, China (2014- 2021)
B.S. in Aquatic Biology, Huazhong Agricultural University, Wuhan, Hubei, China (2010-2014)

II. Research Interests & Expertise

Cancer genomics and transcriptomics; long-read sequencing (Oxford Nanopore and PacBio); computational method development for structural variant analysis; machine learning for genomic sequence analysis; single-cell genomics; RNA-mediated cancer gene regulation; integrative multi-omics.

III. Professional Appointments

Postdoctoral Scholar, Northwestern University, Chicago, IL, USA (2022-Present)

- Developing computational methods for cancer genomics and transcriptomics, with an emphasis on long-read sequencing and complex genomic alterations
- Leading software development for read preprocessing, artifact detection, and multi-platform structural variant analysis
- Collaborating on prostate cancer regulatory genomics, including lncRNA-mediated signaling and transcriptome analysis

Graduate Research Assistant, Huazhong Agricultural University, Wuhan, Hubei, China (2014-2021)

Full-time research position during Ph.D. training, funded by institutional and external grants

- Studied the phylogenetic status and divergence time of myxozoans using phylogenomics
- Investigated the phenotypic, environmental, and genetic adaptation mechanisms of myxozoans using multi-omics approaches

International Intern, Oregon State University, Corvallis, OR, USA (2014)

- Worked on evolutionary genomic and transcriptomic projects of myxozoans under Dr. Jerri Bartholomew
- Investigated cryptic life cycles of myxozoans
- Applied molecular techniques to understand host-parasite interactions

Research Assistant, Huazhong Agricultural University, Wuhan, Hubei, China (2011-2012)

- Studied distribution patterns of stromal eosinophil cells with Dr. Huazhen Liu
- Applied histological and immunohistochemical techniques

IV. Research Contributions

My research focuses on computational methods for analyzing complex genomic and transcriptomic data, particularly from long-read sequencing. My work spans three connected areas: improving the reliability of molecular data, interpreting complex genomic alterations, and investigating RNA-mediated cancer regulation.

1. Improving molecular fidelity in long-read DNA and RNA sequencing

Long-read sequencing can reveal genomic and transcriptomic structures that are difficult to resolve with short reads, but library preparation and amplification can introduce artifacts that resemble genuine molecular events. I develop methods to identify these artifacts while retaining useful sequence information. As lead developer of **DirectClean**, I established a preprocessing and read-rescue workflow for nanopore direct-cDNA sequencing that reduced chimeric alignments in VCaP data from 30.9% to 16.9%. I also contributed to **DeepChopper**, a genomic language model for adapter-associated artifacts in direct RNA sequencing. In ongoing work, I co-developed **ChimeraLM**, a model for identifying whole-genome amplification artifacts and improving structural variant calls in single-cell long-read data.

Selected publications: DirectClean (NAR Genomics and Bioinformatics, 2026); DeepChopper (Nature Communications, 2026); ChimeraLM (manuscript under revision).

2. Resolving complex genomic alterations and their functional consequences

Structural variants are difficult to compare across samples, platforms, and calling algorithms because the same event may have different representations. I co-developed **OctopuSV** and **TentacleSV** to standardize and integrate structural variant calls, support multi-sample comparisons, and automate cross-platform analyses. In related collaborative work, I contributed to **ScanNeo2**, which combines sequence variants, gene fusions, and alternative splicing events for neoantigen prediction. Together, these studies address complementary steps in moving from complex genomic evidence to interpretable variation and potential biological consequences.

Selected publications: OctopuSV and TentacleSV (Bioinformatics, 2025); ScanNeo2 (Bioinformatics, 2023).

3. Investigating RNA-mediated regulation of oncogenic signaling in prostate cancer

In collaborative studies of prostate cancer, I contributed to work on **PRCAT71**, an androgen receptor-regulated lncRNA that interacts with KHSRP to promote AR signaling, and **IGF1R-AS1**, a tumor-specific lncRNA linked to chromatin interactions and MYC signaling. These studies connect transcriptomic analysis with cancer-associated gene regulation.

Selected publications: PRCAT71 (Science Advances, 2025); IGF1R-AS1 (Nature Communications, 2026).

V. Publications

Peer-Reviewed Journal Articles (27 total: 9 first/co-first author; 18 co-authored)

*Qingxiang Guo is shown in bold. † Equal contribution (co-first authorship); * Corresponding author. Symbols are shown for my authorship roles in the cited publications.*

First- and Co-first-Author Publications (9)

1. **Guo Q***, Li Y, Yang R*. DirectClean: artifact-aware preprocessing and read rescue for Oxford Nanopore direct-cDNA transcriptome sequencing. NAR Genomics and Bioinformatics. 2026;8(4):lqag132. doi:10.1093/nargab/lqag132. [Editor's Choice]
2. **Guo Q†***, Li Y†, Wang TY, Ramakrishnan A, Yang R*. OctopuSV and TentacleSV: a one-stop toolkit for multi-sample, cross-platform structural variant comparison and analysis. Bioinformatics. 2025;41(11):btaf599. doi:10.1093/bioinformatics/btaf599.
3. **Guo Q**, Atkinson SD, Xiao B, Zhai Y, Bartholomew JL, Gu Z. A myxozoan genome reveals mosaic evolution in a parasitic cnidarian. BMC Biol. 2022 Feb 18;20(1):51.
4. **Guo Q**, Whipps CM, Zhai Y, Li D, Gu Z. Quantitative insights into the contribution of nematocysts to the adaptive success of cnidarians based on proteomic analysis. Biology (Basel). 2022 Jan 7;11(1).
5. **Guo Q**, Liu Y, Zhai Y, Gu Z. A fast and effective method for dissecting parasitic spores: myxozoans as an example. J Exp Biol. 2020 Jan 1;223(10):jeb214916.
6. **Guo Q**, Li D, Zhai Y, Gu Z. CCPRD: A novel analytical framework for the comprehensive proteomic reference database construction of nonmodel organisms. ACS Omega. 2020 Jun 30;5(25):15370-15384.
7. **Guo Q**, Huang M, Liu Y, Zhang X, Gu Z. Morphological plasticity in *Myxobolus* Bütschli, 1882: a taxonomic dilemma case and renaming of a parasite species of the common carp. Parasit Vectors. 2018 Jul 9;11(1):399.
8. **Guo Q**, Zhai Y, Gu Z, Liu Y. Histopathological and ultrastructural studies of *Myxobolus turpisrotundus* from allogynogenetic gibel carp *Carassius auratus gibelio* in China. Folia Parasitol (Praha). 2016 Oct 14;63.
9. **Guo Q**, Jia L, Qin J, Li H, Gu Z. Myxozoans and our dinner table: pathogenicity studies of *Myxobolus honghuensis* (Myxosporea: Bivalvulida) using a suckling mice model. Foodborne Pathog Dis. 2015 Aug;12(8):653-60.

Co-authored Publications (18)

10. Yang Y, Wang TY, Fry J, Li Y, Meng Q, **Guo Q**, Patchen NE, White KH, Ramakrishnan A, Ren Y, Li Q, Zhang X, Ali T, Dawes C, Fragkogianni S, Irvin P, Alam SK, Hoeppner LH, Zhang X, Yee D, Weiner AB, Schaeffer EM, Liu Y, Zhang X, Dehm SM, Cao Q, Yang R. Tumor-specific lncRNA IGF1R-AS1 trans-regulates chromatin interactions associated with oncogenic MYC signaling. Nature Communications. 2026;17:4171. doi:10.1038/s41467-026-70814-4.
11. Li Y, Wang TY, **Guo Q**, Ren Y, Lu X, Cao Q, Yang R. Genomic language model mitigates chimera artifacts in nanopore direct RNA sequencing. Nature Communications. 2026;17:1864. doi:10.1038/s41467-026-68571-5.
12. Yang Y, Wang TY, Li Q, Lu J, Ren Y, Weiner AB, Fry J, Liu Q, Yum C, Wang R, **Guo Q**, Wan Y, Zhe J, Dong X, Lotan TL, Schaeffer EM, Yang R, Cao Q. Androgen receptor–regulated lncRNA PRCAT71 promotes AR signaling through the interaction with KHSRP in prostate cancer. Sci Adv. 2025 Apr;11(15):eadk6989.

13. Schäfer RA, **Guo Q**, Yang R. ScanNeo2: a comprehensive workflow for neoantigen detection and immunogenicity prediction from diverse genomic and transcriptomic alterations. Bioinformatics. 2023 Nov;39(11):btad659.
14. Zhang B, **Guo Q**, Tu X, Gu Z. Identification of *Myxobolus distalisensis* n. sp. (Cnidaria: Myxozoa) infecting yellow catfish *Tachysurus fulvidraco* (Richardson), with a supplement description of *M. voremkhai* (Akhmerov, 1960) Landsberg and Lom, 1991. Syst Parasitol. 2023 Oct;100(5):473-485.
15. Xiao B, **Guo Q**, Zhai Y, Gu Z. Transcriptomic insights into the diversity and evolution of Myxozoa (Cnidaria, Endocnidozoa) toxin-like proteins. Mar Drugs. 2022 Apr 26;20(5).
16. Tahir UB, **Guo Q**, Gu Z. Fins infestation induced by *Myxobolus xiantaoensis* in yellow catfish *Tachysurus fulvidraco* Richardson, 1846: Some pathophysiological and molecular insights. Microb Pathog. 2021 Apr;153:104772.
17. Wang S, Zhang B, **Guo Q**, Zhai Y, Gu Z. Molecular and light microscopy evidence for the transfer of *Myxobolus honghuensis* from *Carassius auratus gibelio* broodfish to progeny. J Fish Dis. 2020 Oct;43(10):1177-1184.
18. Zhang X, Liu Y, Whipps CM, **Guo Q**, Gu Z. Multiple evolutionary routes of the single polar capsule in *Thelohanellus* species (Myxozoa; Myxobolidae). Int J Parasitol Parasites Wildl. 2019 Apr;8:56-62.
19. Tahir UB, **Guo Q**, Zhao D, Liu Y, Gu Z. Description of *Myxobolus xiantaoensis* n. sp. from the fins of yellow catfish in China: a species previously attributed to *Myxobolus physophilus* Reuss, 1906 in Chinese records. Parasitol Res. 2019 Apr;118(4):1137-1146.
20. Shao J, **Guo Q**, Hu R, Gu Z. Comparative genomic insights into the taxonomy of *Edwardsiella tarda* isolated from different hosts: marine, freshwater and migratory fish. Aquaculture Research. 2018 January;49(1):197-204.
21. Wang Z, Zhou T, **Guo Q**, Gu Z. Description of a new freshwater ciliate *Epistylis wuhanensis* n. sp. (Ciliophora, Peritrichia) from China, with a focus on phylogenetic relationships within family Epistylididae. Journal of Eukaryotic Microbiology. 2017;64(3):394-406.
22. Liu Y, **Guo Q**, Zhai Y, Gu Z. Effects of different preservation methods on morphology of *Myxobolus wulii* spores. Journal of Huazhong Agricultural University. 2016;(4):87-92. (In Chinese with English abstract)
23. Zhai Y, Gu Z, **Guo Q**, Wu Z, Wang H, Liu Y. New type of pathogenicity of *Thelohanellus kitauei* Egusa & Nakajima, 1981 infecting the skin of common carp *Cyprinus carpio* L. Parasitol Int. 2016 Feb;65(1):78-82.
24. Zhai Y, Whipps CM, Gu Z, **Guo Q**, Wu Z, Wang H, Liu Y. Intraspecific morphometric variation in myxosporeans. Folia Parasitol (Praha). 2016 Apr 1;63.
25. Deng Q, **Guo Q**, Zhai Y, Wang Z, Gu Z. First record of *Chilodonella piscicola* (Ciliophora: Chilodonellidae) from two endangered fishes, *Schizothorax o'connori* and *Oxygymnocypris stewartii* in Tibet. Parasitol Res. 2015 Apr;114(8):3097-3103.
26. Jia L, Gu Z, Liu Y, **Guo Q**, Li D, Zhai Y. Preparation and characterization of polyclonal antibodies against *Myxobolus honghuensis* (Myxosporea: Bivalvulida). Journal of Huazhong Agricultural University. 2015;(6):113-118. (In Chinese with English abstract)
27. Zhang JY, Gu ZM, Kalavati C, Costa Eiras J, Liu Y, **Guo Q**, Molnár K. Synopsis of the species of *Thelohanellus* Kudo, 1933 (Myxozoa: Myxosporea: Bivalvulida). Syst Parasitol. 2013 Nov;86(3):235-56.

Manuscripts Under Revision

Li Y†, **Guo Q†**, Yang R. ChimeraLM: a language model enables accurate structural variant detection in whole-genome amplified long-read sequencing. Manuscript under revision.

VI. Patents

1. Gu Z, **Guo Q**, Zhai Y, Liu Y. A method to isolate and purify the nematocyst of *Myxobolus honghuensis* (Myxosporea: Bivalvulida). China Patent Application No. CN110055181B. 2022.
2. Gu Z, Li D, Liu Y, **Guo Q**. A PCR method for simultaneous detection of multiple myxozoans in allogynogenetic gibel carp *Carassius auratus gibelio*. China Patent No. ZL201610051386.8. 2019.
3. Gu Z, Jia L, Liu Y, Zhai Y, Yuan J, Qin J, Li D, **Guo Q**. Preparation and application of polyclonal antibodies against the shell valves of *Myxobolus honghuensis* (Myxosporea: Bivalvulida). China Patent No. CN103755806B. 2016.
4. Gu Z, Jia L, Liu Y, Zhai Y, Yuan J, Qin J, Li D, **Guo Q**. Preparation and application of polyclonal antibodies against the polar filaments of *Myxobolus honghuensis* (Myxosporea: Bivalvulida). China Patent No. CN103880954B. 2015.
5. Gu Z, **Guo Q**, Zhai Y, Liu Y. A method to isolate and purify the shell valves of *Myxobolus wulii* (Myxosporea: Bivalvulida). China Patent Application No. ZL CN110004104B. 2022

VII. Presentations

Oral Presentations

1. Resolving chimeric artifacts across Nanopore DNA and RNA sequencing. Oxford Nanopore User Group Meeting, Chicago, IL, USA (September 29, 2026). Invited presentation.
2. Paving the way to structural variant analysis in prostate cancer with nanopore long-read sequencing. Forum on Biochemistry, Epigenetics, and Metabolism (BEaM), Simpson Querrey Institute for Epigenetics, Northwestern University, Chicago, IL, USA (October 20, 2023).
3. Morphological plasticity in *Myxobolus*: a taxonomic dilemma case. The 3rd Chinese Protozoological Society Scholars Symposium, Jilin, China (2018).
4. Lights into the evolutionary scenario of extrusion apparatus in eukaryotes. The 19th Symposium of Chinese Protozoological Society, Sun Yat-sen University, China (2017).
5. Histopathological and ultrastructural studies of *Myxobolus turpisrotundus*. The 2nd Chinese Protozoological Society Scholars Symposium, Shanghai, China (2016).
6. Pathogenicity studies of *Myxobolus* using a suckling mice model. The 7th International Symposium on Aquatic Animal Health, Portland, OR, USA (2014).
7. Pathogenicity studies of *Myxobolus* using a suckling mice model. The 17th Symposium of Chinese Protozoological Society, Jilin University, China (2013).

Poster Presentations

1. Lights into the evolutionary scenario of extrusion apparatus in eukaryotes. The annual meeting of the Society for Molecular Biology and Evolution, Austin, USA (2017)
2. Comparative transcriptomic insights into the origin and evolution of myxozoans. The 3rd National Academic Summit on Functional Genomics, Beijing, China (2016)
3. Comparative transcriptomic insights into the origin and evolution of myxozoans. The 18th Symposium of Chinese Protozoological Society, Yantai, China (2015)

VIII. Honors and Awards

1. Second Prize, Simulated International Academic Conference of Postgraduate Students (2016)
    - Only two prize winners in Huazhong Agricultural University this year
2. The Best Presenter Award, Academic Annual Meeting of College of Fisheries, Huazhong Agricultural University (2015)
3. Champion Award, The First Frasergen Bioinformatics Contest (2014)
    - Only one prize winner out of the 150 contestants from universities in China
4. The Outstanding Individual, The Professional Social Practice Competition in Hubei Province (2012)
    - Awarded for the best team leader in professional practice; only 415 winners in China this year

IX. Teaching and Mentoring Experience

Teaching Assistant, Fish Diseases course, Huazhong Agricultural University (2015)

- Led section discussions for 31 students
- Prepared laboratory reagents
- Wrote and graded exams

Undergraduate Mentor, Fisheries, Huazhong Agricultural University (2012-2013)

- Provided social support, professional development, and career guidance to 35 first-year students

Supervisor, Huazhong Agricultural University (2015-2018)

- Direct supervisor of seven undergraduates on their senior theses

Research Mentor, Northwestern University (2023)

- Mentored Timothy Pan, PhD student (Driskill Graduate Program), during laboratory rotation
- Provided guidance in experimental design, data analysis, and lab workflows

Research Mentor, Northwestern University (2024)

- Supervised Alice Kong, high school student (Payton College Preparatory High School), ResearchSTAR Summer Intern
- Introduced core research concepts, laboratory techniques, and scientific thinking

X. Professional Service

Manuscript Review

Peer reviewer for 56 individual review assignments across 23 journals (2016–2026): International Journal of Molecular Sciences (9); Cancers (7); Frontiers in Bioinformatics (7); Foodborne Pathogens

and Disease (4); Computational and Structural Biotechnology Journal (4); Journal of Clinical Medicine (3); Wellcome Open Research (3); Biology (2); BMC Genomics (2); Journal of Imaging (2); Antibodies (1); Antioxidants (1); BioMedInformatics (1); Biomedicines (1); BMC Biology (1); BMC Genomic Data (1); Cells (1); Diversity (1); Genes (1); GigaByte (1); Journal of Marine Science and Engineering (1); Scientific Reports (1); Translational Oncology (1).

Professional Memberships

- Society for Molecular Biology and Evolution (2017-Present)
- China Society of Fisheries (2015-2020)
- Chinese Protozoological Society (2013-2020)

XI. Technical Skills

Bioinformatics and Omics Analysis

- Long-read sequencing (Oxford Nanopore and PacBio): read preprocessing, alignment, transcriptome analysis, and quality control
- Structural variant calling, harmonization, comparison, annotation, and interpretation
- Cancer transcriptomics: gene fusions, alternative splicing, lncRNAs, and RNA-associated regulation
- Single-cell genomics and transcriptomics; whole-genome amplification artifacts; multi-omics integration
- Genome assembly, annotation, comparative genomics, and proteomics (including CCPRD)

Computational Skills

- Programming: Python, R, Bash/Shell; additional experience with Perl and SQL
- Computing and reproducibility: Linux, SLURM/HPC, Docker, Apptainer, Git, Conda/Mamba, Snakemake
- Machine learning and data analysis: genomic language models, sequence classification, CNN and Mamba-based architectures
- Additional experience: phylogenomics and evolutionary analysis (OrthoMCL, RAxML, IQ-TREE, PhyloBayes, BEAST, R8S, HYPHY, PAML)

Experimental and Laboratory Skills

- Oxford Nanopore long-read library preparation and sequencing
- Molecular biology: PCR, Western blotting, cloning, and related laboratory procedures
- Animal handling (mice and fish); histology and immunohistochemistry
- Microscopy (light, fluorescence, and transmission electron microscopy)